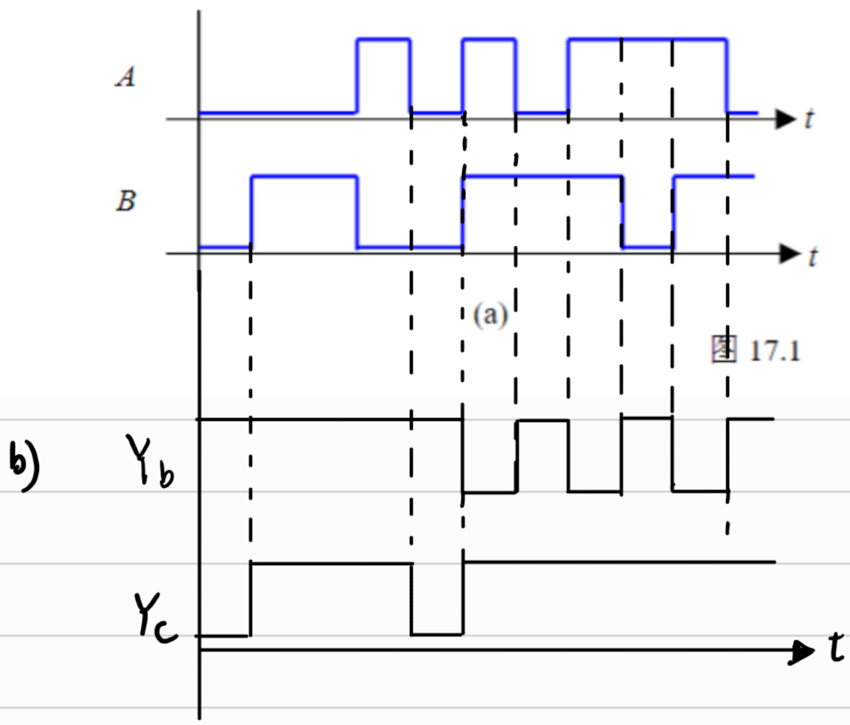
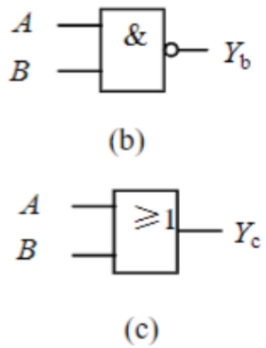
17.1 （门电路功能）作用于各门电路输入端的信号波形如图 17.1（a）所示。画出图 17.1（b）（c）各电路输出端 *Y* 的波形图。



(a)

图 17.1



17.2 （逻辑代数化简）Using Boolean algebra（布尔代数，逻辑代数） techniques, simplify the following expression as much as possible, and implement the logic circuit using gate circuits.

(a) $X=(AB+A\overline{B}+\overline{A}B)(A+B+D+\overline{A}\overline{B}\overline{D})$

(b) $X=\overline{A}\overline{B}C+\overline{A+B+\overline{C}}+\overline{A}\overline{B}\overline{C}D$

a) $X=[A(B+\bar B)+\bar A B](A+B+D+\bar A\bar B\bar D)$

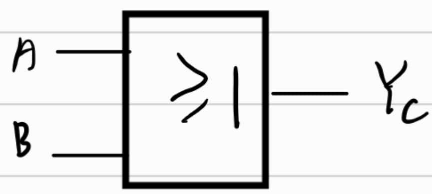
$=(A+\bar A B)(A+B+D+\bar A\bar B\bar D)$

$=(A+B)(A+B+D+\bar A\bar B\bar D)$

$=(A+B)(A+B+D+\bar A+\bar B+\bar D)$

$=(A+B)(A+\bar A+B+\bar B+D+\bar D)$

$=A+B$



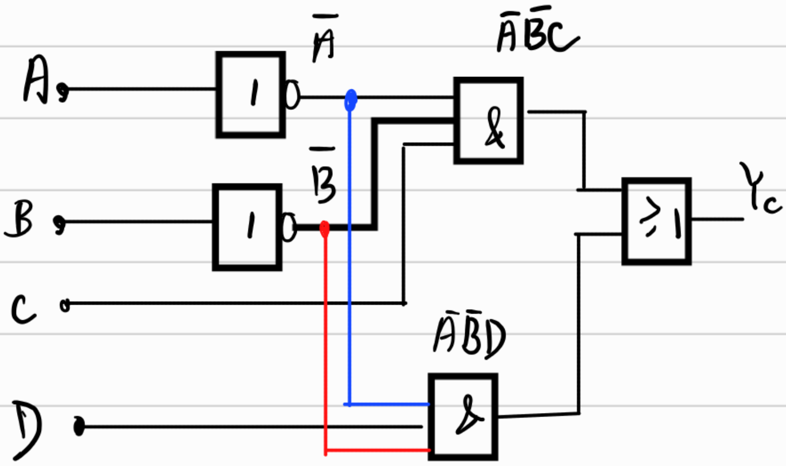
b) $X=\bar A\bar B C+\bar A\cdot\bar B C+\bar A\bar B\bar C D$

$=\bar A\bar B C+\bar A\bar B\bar C D$

$=\bar A\bar B(C+\bar C D)$

$=\bar A\bar B(C+D)$

$=\bar A\bar B C+\bar A\bar B D$



17.3 （卡诺图化简）Use a Karnaugh map to simplify the following expression to a minimum sum-of-product form (积和式，与或式), and implement the logic circuit using NAND gates (与非门). $A\bar{B}(C+\bar{C})$ $CD(A+\bar{A})$

(a) $Y = A\bar{B} + A\bar{B}\bar{C}\bar{D} + CD + B\bar{C}D + ABCD$

(b) $Y = AC + ABD + BC + BD$

a)

AB \ CD	00	01	11	10
00	0	0	1	0
01	0	1	1	0
11	0	1	1	0
10	1	1	1	1

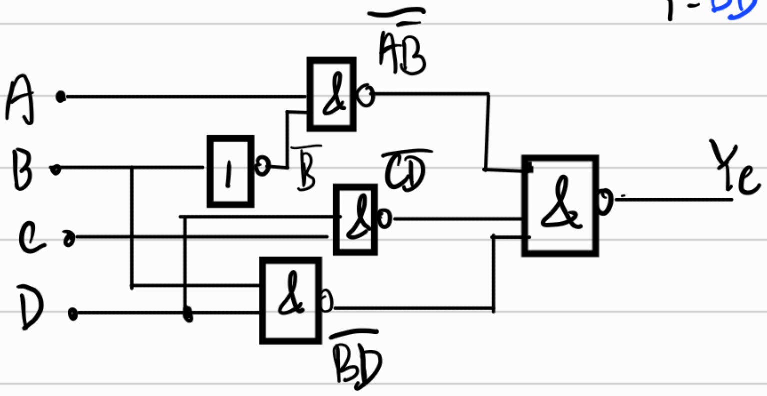
$A\bar{B}C + A\bar{B}\bar{C} + ACD + \bar{A}CD$

$A\bar{B}C(D+\bar{D}) + A\bar{B}\bar{C}(D+\bar{D}) + ACD(B+\bar{B}) + \bar{A}CD(B+\bar{B})$

$= A\bar{B}CD + A\bar{B}C\bar{D} + A\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D} + ABCD + A\bar{B}CD + \bar{A}BCD + \bar{A}\bar{B}CD$

$B\bar{C}D(A+\bar{A}) = AB\bar{C}D + \bar{A}B\bar{C}D$

$Y = BD + CD + A\bar{B} = \overline{\overline{BD}\cdot\overline{A\bar{B}}\cdot\overline{CD}}$



b)

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	0	1	1	1
10	0	0	1	1

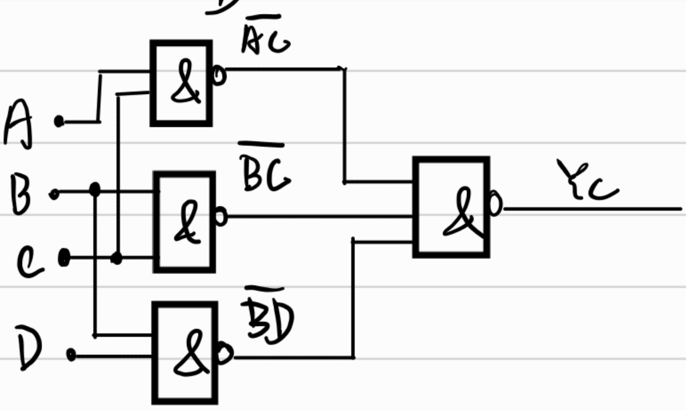
BC

BD

AC

$Y = BC + AC + BD$

$= \overline{\overline{BC}\cdot\overline{AC}\cdot\overline{BD}}$



17.4 （卡诺图化简）利用卡诺图化简

(a) $F(A,B,C)=\Sigma(2,3,4,6)$

(b) $F(A,B,C,D)=\Sigma(0,1,2,3,4,6,8,9,10,11,12,14)$

a)

A \ BC	00	01	11	10
0	0	0	1	1
1	1	0	0	1

$\bar{A}B$ $\quad A\bar{C}$ $\quad Y=\bar{A}B+A\bar{C}$

b)

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	0	1
11	1	0	0	1
10	1	1	1	1

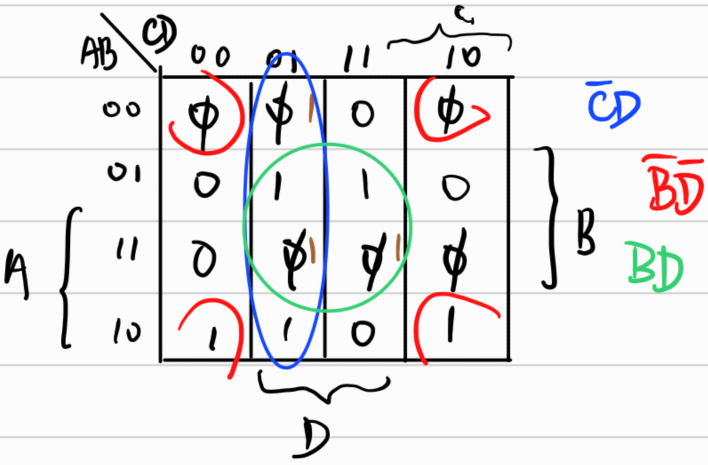
$\bar{D}$ $\quad \bar{B}$

$Y=\bar{D}+\bar{B}=\overline{B\cdot D}$

17.5 （卡诺图化简、逻辑代数化简综合题）已知逻辑函数 F 的真值表如图 17.5 所示（未给出的项为任意项），先用卡诺图化简法并再进一步用逻辑代数化简法写出 F 的最简表达式，使 F 可以用尽量少的门电路实现。

A B C D	*F*
0 0 1 1	0
0 1 0 0	0
0 1 0 1	1
0 1 1 1	1
0 1 1 0	0
1 0 0 0	1
1 0 0 1	1
1 0 1 0	1
1 0 1 1	0
1 1 0 0	0

图 17.5



$F = \overline{C}D + \overline{B}\,\overline{D} + BD$